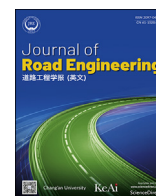




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Review Article

Review on composition properties, functionalization of bio-oil and its rejuvenation behavior and mechanism on aged asphalt

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HIGHLIGHTS

- Bio-oil has good diffusivity, similar to the components of petroleum asphalt.
- Functionalization methods of bio-oil include purification, composition modification and component conversion.
- The active groups and derivatives in bio-oil can be converted into the missing components of aged asphalt binders.
- Both dissolving and diluting rejuvenation mechanisms depend on the type of bio-oil and aged binder.

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ABSTRACT

The objective of this paper is to comprehensively review the research progress of bio-oil properties and hot rejuvenation behavior and mechanism to aged asphalt. The preparation process, composition characteristics of bio-oils and their component correspondence with petroleum asphalt were compared. The diffusion and fusion effects of various bio-oils in aged asphalt were introduced. Bio-oil cannot be used as a direct alternative of petroleum asphalt, but it has the potential to effectively rejuvenate aged asphalt binders due to the component similarity with petroleum asphalt and good diffusion properties. For the asphalt rejuvenation, the functionalization treatment methods of bio-oil were discussed such as purification, composition modification and component conversion. The active groups and derivatives in bio-oil can be converted into the missing components of the aged binder through phenolate, grafting, polycondensation, resinifying, but the conversion process and mechanism are still unclear. From the perspectives of diffusion behavior, components regulation, dissolving asphaltene and micro-rejuvenation effect, the rejuvenation behavior and mechanism of bio-oil on aged asphalt were elaborated, and the effects of various types of bio-oil and aged asphalt on rejuvenation behavior were analyzed. The preparation process and dosage of bio-rejuvenator were summarized. The rejuvenation effects of bio-oil on aged asphalt were comprehensively investigated from the aspects of high and low temperature performances, rheological properties, microstructure and chemical composition of bio-rejuvenated asphalt binders. Finally, the limitations of bio-oil used as asphalt rejuvenators were discussed, and future research directions were prospected, which can provide reference and theoretical basis for the development of high-performance bio-oil rejuvenating agents and the engineering application of bio-oil to improve the properties of aged asphalt materials.

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1. Introduction

Asphalt mixture is widely used as paving materials for high-grade highway pavement, airport pavement and other transportation infrastructures due to the good service performance. However, asphalt mixture suffers from the comprehensive action of traffic loads and various weather conditions, which will lead to the deterioration of performance until it is difficult to meet the service requirements. It is estimated that about 12% of asphalt pavements need to be maintained and repaired in a large area each year in China, and the asphalt pavements, which are hard to continue to service, need to be re-paved. It is reported that the amount of reclaimed asphalt pavement (RAP) materials produced by the removal of old asphalt pavements reaches 50–80 million tons per year, and increases at a rate of 15% year by year. Meanwhile, Petroleum refers to a non-renewable resource, and road materials derived from petroleum such as asphalt, modifiers, and rejuvenators are increasingly shortage. Therefore, how to achieve the green and efficient reclamation of asphalt mixture, alleviate resources shortage, and reduce environmental problems of waste has become a common concern of the transportation and environmental protection departments.

Under the coupling action of various weather conditions (heat, oxygen, light, water, etc.), asphalt is constantly aging and becoming harder and brittle, which leads to a significant decline in the mechanical properties of the mixture. The RAP recycled by milling can be used in new paving applications by hot mix, warm mix, cold mix, and other processes, which can significantly save production costs and reduce the environmental pollution in the construction of asphalt pavement (Osman et al., 2024; Zauamanis et al., 2014). Among them, hot recycling technology is one of the important ways of asphalt pavement rejuvenation, which can give full play to the action of the binder and ensure the performance of recycled asphalt pavement. Compared with the virgin asphalt, the aging of asphalt binder in RAP mainly manifested in two aspects. The one is the volatilization of light components such as saturates and aromatics, as well as the selective adsorption of light components by aggregates, lead to the change of asphalt components and mass loss (Loise et al., 2019). The other is the chemical aging, mainly based on the thermal oxidation or photo-oxidation reaction, lead to the structural change of the molecular chain of composition, and the component imbalance and types of colloidal structure may change as a result of these process (Behnood et al., 2016). It can be seen that the imbalance of compositions and colloidal structure is the main manifestation of aging and also the fundamental reasons of performance degradation. In the hot recycling process of RAP, it is generally necessary to add a certain amount of virgin asphalt or rejuvenator according to the aging degree to restore the rheological properties and technical indicators of aged asphalt binder (Xu et al., 2021). Due to the limitation of the process of hot recycling, mixing time and other factors, there are several requirements for rejuvenators. The first is good dissolution and penetration effect, and ability to effectively dissolve and disperse the asphaltene. The second is the ability to produce synergistic effects with asphalt binder to make rejuvenated asphalt have good rheological properties. The last is good heat resistance and anti-aging properties to ensure the service durability of recycled asphalt pavement.

In recent years, many scholars in the field of road and materials have been committed to the study of rejuvenators with desired properties, and have achieved many results. At present, there are two main types of asphalt rejuvenators. The one type is traditional rejuvenators derived from ethylene monomer, a raw material of the petrochemical industry. These agents are hydrocarbon light oils prepared by organic synthesis or industrial waste oil separation process and largely rely on the petroleum industry. The traditional rejuvenators have good affinity with asphalt, and have good dissolving and penetrating ability, but they tend to be oxidized at high temperature, and the proportion of effective components such as alkanes and aromatics hydrocarbons is difficult to control. Therefore, this type of rejuvenator has a large variability in properties. And the other type is bio-oil rejuvenator derived from animals and plants,

such as waste cooking oil, vegetable oil, animal manure, and bio-oil prepared by high temperature carbonization of wood, etc (Cavalli et al., 2024; Raouf and Williams, 2010; Samieadel et al., 2020; Yan et al., 2022; Zhang et al., 2019a, 2022). Yu et al. (2014) adopted waste vegetable oil derived from fast and convenience food frying oil to prepared bio-rejuvenator. Cai et al. (2019) developed bio-rejuvenator by using plasticizer and cotton oil. This type of rejuvenators has a wide range of raw material sources, which can effectively utilize domestic waste resources and reduce the dependence on petrochemical resources. However, there still exists some problems in the research on bio-rejuvenator as follows.

- (1) Compared with petrochemical materials, bio-oil has poor stability and high water content, thus exhibiting a poor rejuvenating efficiency of aged asphalt binder.
- (2) Various types of bio-oil with complex composition need to further clarify the rejuvenation mechanism to improve the rejuvenation effect of bio-oil.
- (3) The component design and process conditions of bio-rejuvenated asphalt are still immature, which largely limits the development of bio-rejuvenators.

This paper reviews the research achievements and progress on the characteristics of bio-oil and the application in hot recycling of aged asphalt materials. On the basis of reviewing preparation process and composition characteristics of bio-oil, the composition and components correspondence bio-oil and asphalt was analyzed, and the potential of bio oil for asphalt rejuvenation was discussed. And then, the functionalization treatment methods of bio-oil for asphalt rejuvenation were summarized. The rejuvenation behavior and action mechanism were elaborated on from different perspectives of diffusion behavior, component regulation, asphaltene depolymerization, and microscopic changes. In addition, the rejuvenation effect of bio-oil on aged asphalt and the performance of rejuvenated asphalt were evaluated, which can provide theoretical basis and reference for the research and development of bio-rejuvenator and the application of bio-rejuvenated asphalt. Fig. 1 shows the structure of this work in the form of a flowchart.

2. Preparation process and composition properties of bio-oil

2.1. Preparation process and conditions of bio-oil

Bio-oil mainly is derived from various organic matter produced by plant photosynthesis, which is a mixture of many organic matter degradation products, such as plant straw, animal and plant waste, microorganisms, waste oil, etc. At present, the production methods of bio-oil can be divided into three pathways: physical processing, biochemistry conversion, and thermochemical conversion. The physical or mechanical processing squeezes bio-oils by applying pressure, which is a low technology costs and yields method, such as peanut oil, rapeseed oil, soybean oil, corn oil, and other fresh extracted oils (Zahoor et al., 2021). The biochemical conversion mainly includes anaerobic fermentation and biological enzyme technologies, with high cost, slow conversion rate, and low yield (Mishra et al., 2020). Fresh oil obtained by squeezing technology is usually used in catering industry, and the biochemical conversion method has been widely used in the energy industry. However, thermochemical technology provides a sustainable solution for the application of waste biomass materials in the field of road materials.

Thermochemical conversion methods mainly involve pyrolysis and liquefaction. The pyrolysis technology refers to the process that biomass materials are converted from solid state into combustible gas, bio-oil, char, and other small molecular products performed in hypoxic or the absence of air conditions (Fini et al., 2011). The proportion of three products depends on the process parameters. The preparation process of pyrolysis is generally as follows: the temperature is 500 °C–650 °C, gas residence time is 1–2 s, and heating rate is 10–200 °C/s (Zahoor et al.,

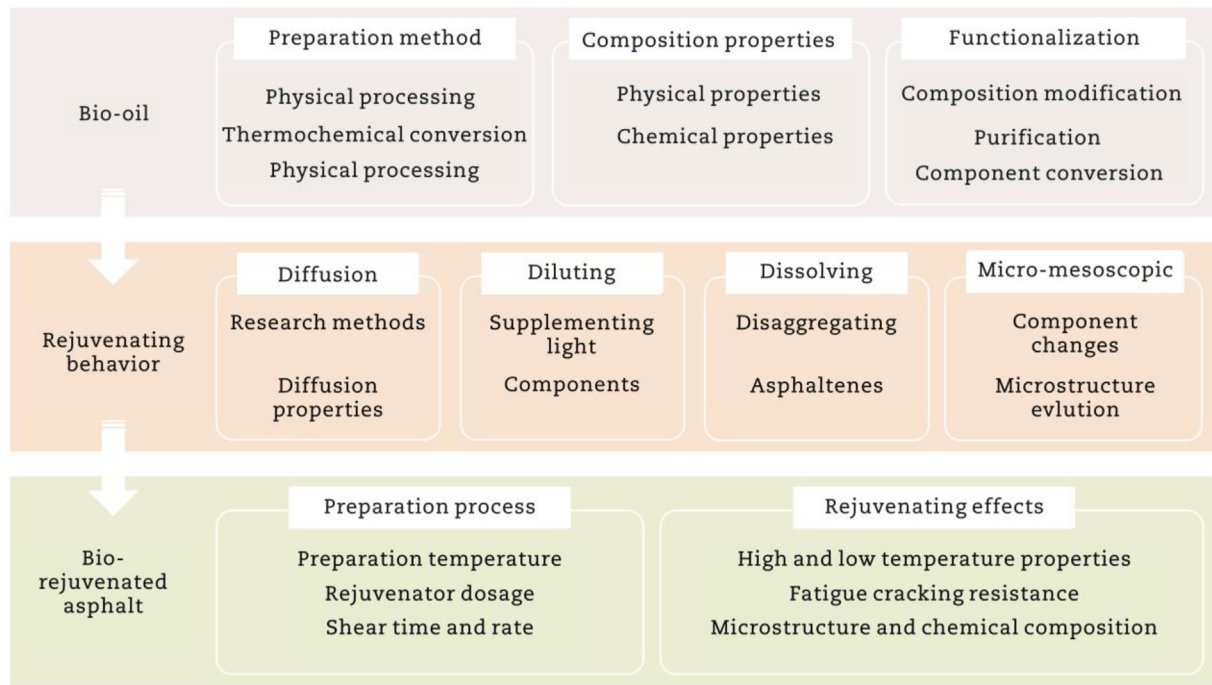


Fig. 1. Structure of the review paper.

2021). The fast pyrolysis process is showed in Fig. 2. Thermochemical liquefaction technology is a process in which biomass is degraded from a solid state to low molecular weight bio-oil in solvents, as shown in Fig. 3. The liquefaction solvents usually include water, phenol, and polyol systems (Jiang et al., 2021), and the solvent can improve the heat transfer efficiency and uniformity, and participate in the depolymerization of biomass components, the decomposition of monomer, and the recombination of reaction intermediates. Therefore, the physical and chemical properties of bio-oil are related with the liquefaction solvent type used in liquefaction process (Basar et al., 2021). Table 1 shows the thermochemical preparation process of biomass. The yield and characteristics of bio-oil are affected by many factors, such as biomass type and particle size, solvents, catalyst type and concentration, liquid-solid ratio, liquefaction time, pressure, temperature and so on (Jiang et al., 2018). The mixed liquefaction solvent system usually tends to have a higher liquefied yield than the single solvent. At the same time, the pretreatment of biomass materials can enhance the yield of bio-oil. Generally, there are three pretreatment methods. (1) Physical pretreatment, such as chopping, grinding, etc. (2) Chemical pretreatment, such as using acid or alkali solvents to unlock the lignocellulosic structure. (3) Biological

pretreatment, such as biological enzyme catalyzing (Song et al., 2020).

Waste biomass materials can be classified into three types: (1) wood fiber, containing crop straw, waste wood and forest waste; (2) waste oil, containing hogwash oil and refined oil residues; and (3) animal manure. The preparation processes of bio-oil derived from lignocellulosic biomass and animal manure are relatively mature, while there are various pathways to obtain bio-oil from animal fats and refined oil residues. Nciri et al. (2020) cut waste the pork fat into small pieces, heated and stirred at 110 °C for 1 h to prepared bio-oil. After filtration, anhydrous sodium sulfate was used to remove water in the product.

In summary, the biomass sources and production methods of bio-oil are myriad. The physical or mechanical processing, a low technology costs and yields method, is usually used in catering industry. The biochemical conversion method is mainly used in the energy industry, with high cost, slow conversion rate, and low yield. Compared with physical processing and biochemistry conversion, the thermochemical conversion method provides a sustainable solution for the application of waste biomass materials in the field of road materials. The hydrothermal liquefaction process is usually performed in sub/supercritical water, with the high cost of equipment, harsh reaction conditions, low yields and

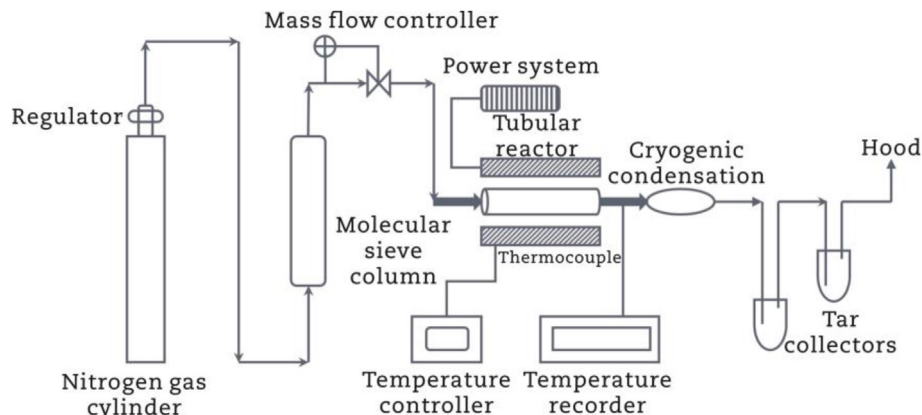


Fig. 2. Schematic diagram of fast pyrolysis process (Tsai et al., 2007).

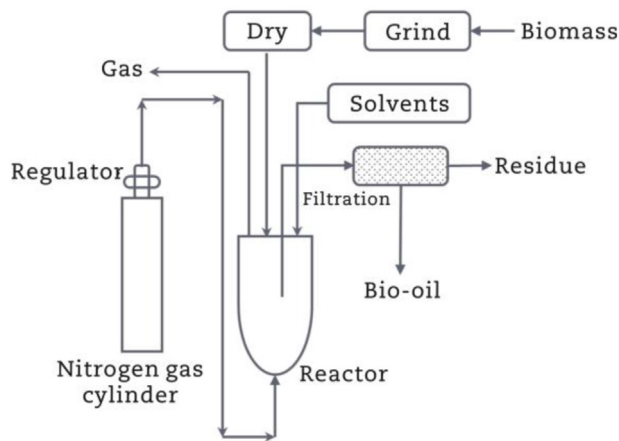


Fig. 3. Schematic diagram of thermochemical liquefaction process.

high oxygen content. Moreover, the biomass with high moisture content can negatively affect the pyrolysis process (Akhtar and Saidina Amin, 2012). The liquefaction technology in organic solvent requires simple equipment and mild reaction conditions. The oxygen content of liquid products is lower, and the composition of the products is easy to control (Jiang et al., 2021, 2022a). However, the following problems of liquefaction should not be ignored. (1) The recovery rate of solvent is low. (2) The use of solvents may lead to environmental pollution. (3) Liquefaction technology in pure organic solvents requires pre-drying process of biomass materials, which may increase costs.

At present, the preparation technology of bio-oil derived from lignocellulosic biomass and animal manure has been relatively mature. While, the applicability of pyrolysis and liquefaction technology in solvent for the preparation of waste oil, the selection of liquefaction solvents and the optimization of liquefaction conditions require further study. It is necessary to aim at environmentally friendly solvent liquefaction

technology, reduce the amount of solvent and make it participate in the subsequent strengthening reactions to enhance the properties of rejuvenators, thereby avoiding solvent recovery and simplifying the process.

2.2. Component correspondence between bio-oil and asphalt

When used for asphalt rejuvenation, the compatibility of bio-rejuvenator with the aged binder is a crucial factor affecting the rejuvenation effect. Based on the compatibility theory, the physical and chemical properties and composition of bio-oil requires good similarity with aged asphalt. Due to a wide range of sources and complex ingredients of biomass materials, there are various phy-chemical reactions during the conversion process of bio-oil. Therefore, the composition and physicochemical properties of bio-oil are much complicated. In general, bio-oil is a dark brown liquid or semi-liquid, and the solid content of bio-oil is less than 1 wt%. The ash content is less than 0.2 wt%, and bio-oil has a high moisture content, some even up to 30 wt% (Zahoor et al., 2021; Zhang et al., 2021a). The high water content has a significant effect on the phase stability of bio-oil (Oasmaa et al., 2015). The density of bio-oil ranges from 0.9 to 1.3 g/cm³, which is close to the density of base asphalt, and the viscosity is lower than that of petroleum asphalt. Bio-oil has the potential to dissolve the asphaltene and reduce the viscosity of the aged asphalt. Bio-oil is of acidic nature and its pH is between 2 and 4. But the bio-oil, derived from swine manure (Hosseinnazhad et al., 2015), biodiesel residues (Sun et al., 2016), and rubber seeds (Onoji et al., 2016), has less acidic and its pH can be close to 6.

The majority of the raw materials for the preparation of bio-oil are woody materials such as crop straw and sawdust, which contain the high content of cellulose, hemicellulose and lignin. Therefore, the bio-oil is mainly composed of five elements of C, H, O, N, and S, which is basically consistent with the element composition of petroleum asphalt. While, some bio-oil contain trace content of Si, P, Ca and others (Huber and Dumesic, 2006). The elemental compositions of bio-oils for different sources are presented in Table 2.

Table 1
Thermochemical preparation of bio-oils from different sources.

Source	Reference	Preparation method	Oil yield (%)	Preparation process
Swine manure	Fini et al. (2011)	Pyrolysis	–	305 °C, 10.3 MPa, 80 min, cooled to room temperature, vacuum filtration
Rice husk	Tsai et al. (2007)	Fast pyrolysis	40.0	500 °C, 200 °C/min, residence time 1 min, flow rate of N ₂ 1000 cm ³ /min, 10 °C condensation
Hogwash oil	Zheng et al. (2010)	Catalytic pyrolysis	77.9	540 °C, 70 min, mass ratio of composite catalyst to oil was 1:300
Pine wood	Brand et al. (2013)	Supercritical liquefaction	65.8	Ethanol was adopted as the solvent, solid-liquid ratio 0.06, 400 °C, 120 min
Wheat straw	Seehar et al. (2020)	Subcritical liquefaction	32.3	350 °C, 15 min, 18 MPa, 20% straw, 2% K ₂ CO ₃
Pine sawdust	Cheng et al. (2010)	Hydrothermal liquefaction	65.0	300 °C, 15 min, 50 wt% water-ethanol or water-methanol liquefaction, liquid-solid ratio 10:1
Rice straw	Jiang et al. (2021)	Phenol liquefaction	88.0	90 min, 140 °C, 2.0 MPa liquid-solid ratio 4:1, 2.5% sulfuric acid
Corn straw	Zhang et al. (2019b)	Polyol liquefaction	98.6	60 min, 170 °C, 338 r/min, diethylene glycol and 1,2-propanediol (6:1), 10% concentrated phosphoric acid, liquid-solid ratio 5:1
Poplar sawdust	Chu et al. (2016)	Solvent liquefaction	93.0	135 °C, 120 min, 70% phenol and 30% PEG 400, liquid-solid ratio 1.5, 4% concentrated sulfuric acid (H ₂ SO ₄ , 98%)

Table 2
Elemental composition of bio-oils from different sources.

Biomass	C (%)	H (%)	O (%)	N (%)	S (%)	Reference
Petroleum asphalt	85.54	9.91	0.91	0.43	3.21	Bao et al. (2020)
Waste vegetable oil	77.77	10.31	10.26	0.37	0.17	Cao et al. (2018)
Rapeseed	62.10	9.10	24.90	3.90	–	Onay and Koçkar (2006)
Rice husk	42.00	5.40	39.30	0.40	–	Alvarez et al. (2014)
Cotton straw	42.30	7.90	49.40	0.30	0.20	Zheng et al. (2008)
Rice straw	43.25	5.62	48.80	2.11	0.22	Jung et al. (2008)
Bamboo sawdust	46.90	5.85	47.02	0.21	0.02	Jung et al. (2008)
Soybean straw	43.58	5.24	44.72	2.52	–	Wang et al. (2018b)
Swine manure	72.58	9.76	13.19	4.47	–	Fini et al. (2011)

Table 2 shows that the chemical composition of bio-oil significantly varies with biomass materials. Among them, the N content of the products derived from swine manure and soybean are higher in comparison to the other bio-oils. This is mainly because these biomass sources have higher N content (Yang et al., 2015). Younas et al. (2017) utilized gas chromatography-mass spectrometry (GC-MS) to analyze the compounds the bio-oil generated from rice straw, and found that this bio-oil mainly contained ketones, esters, benzenes, phenols, oxygenated compounds, carboxylic acids, and hydrocarbons. Gong et al. (2016) found that the bio-oil extracted from waste oil was a multi-component mixture including seven kinds of long-chain fatty acid methyl esters. Waste edible oil and soybean oil contain fatty acids such as oleic acid, palmitic acid, and linoleic acid (Ahmed and Hossain, 2020). The bio-oil produced by swine manure has strong polarity and the high content of polar functional groups such as olefins and amides groups, which can improve the moisture resistance of asphalt (Fini et al., 2011). In addition, the proportion of components of bio-oil varies with the preparation process. Zhang et al. (2019b) adopted diethylene glycol (DEG) with propanediol glycol (PG) and ethylene glycol (EG) with PG as liquefying agents respectively, and characterized the liquefied bio-oil derived from corn straw by GC-MS. The alcohols in the two bio-oils are 37.75% and 83.14%, respectively, the organic acids are 37.75% and 15.38%. It can be seen that bio-oil is an organic mixture based on high molecular weight organic matter and complex compounds.

In general, bio-oil is rich in functional groups such as alkanes and cycloalkanes, which are similar to the aromatic fraction of petroleum asphalt (Zhang et al., 2020). According to the chemical fractions analysis method of petroleum asphalt, the difference of SARA fractions (saturates, aromatic, resins and asphaltenes) between conventional asphalt and bio-oils obtained from various sources is shown in Fig. 4.

As seen from Fig. 4, the categories of components are consistent, although there is great difference in the ratio of SARA between bio-oil and petroleum asphalt. Among them, the bio-oil produced from swine manure has the high content of aromatics and resins, which can effectively dissolve the asphaltenes in aged asphalt. However, some researchers have pointed out that there is a significant difference in the structure of asphaltenes between bio-oil and petroleum asphalt (Pahlan et al., 2018b; Xu et al., 2014). The aromatics content of waste cooking oil is very high, while the content of asphaltenes and resins is much lower than that of petroleum asphalt. From the distribution of SARA fractions, the bio-oil generated by waste wood is closer to petroleum asphalt, but it contains substances insoluble in toluene, which may be solid residues in the preparation process of bio-oil.

From the results of elemental composition and SARA fractions analysis, bio-oil and petroleum asphalt have a preferable correspondence, but

the potential compatibility of them is difficult to quantitatively analyze. Yang et al. (2017) investigated the compatibility between bio-oil extracted by waste wood and petroleum asphalt by automated flocculation titrimetry (AFT), and found that bio-oil and petroleum asphalt can form a stable and compatible blend system. But the compatibility decreases with the increase in bio-oil content. Based on molecular dynamics simulation, Zhang et al. (2010) analyzed the effect of rejuvenator components on the compatibility with aged asphalt binder. It indicates that aromatics can effectively improve the compatibility of the rejuvenator with aged binder, while saturates would adversely affect the compatibility. And the higher in dosage, the greater for the impact. Therefore, it is considered that the reason for the decrease of the compatibility of bio-asphalt with high-content bio-oil may be caused by the high content of saturates in the bio-oil.

From the perspective of physical properties, elemental composition and component analysis, it is generally concluded that bio-oil cannot be used as a direct substitute for petroleum asphalt. However, the majority of bio-oil has the characteristics of low viscosity and high content of light components, and the components similarity between bio-oil and petroleum bitumen ensures better compatibility with each other. Thus the bio-oil has a great potentiality in rejuvenating aged asphalt. It is worth noting that the high aromatics content leads to strong acidity of bio-oil, which increases the difficulty of storage and transportation. Compared with petrochemical materials, bio-oil has poor stability and high water content, which has an adverse effect on the bio-rejuvenated asphalt. The molecular weight range of bio-oil is narrow, and is easy to volatilize and degrade, which will lead to a high temperature sensitivity and poor low temperature and anti-aging properties of rejuvenated asphalt. Therefore, it is necessary to carry out further technical treatment and improvement used as bio-rejuvenator.

2.3. Functionalization treatment for bio-oil for asphalt regeneration

Bio-oil contains water, hydroxyl, carbonyl compounds and other volatile substances, which affect thermal stability and aging resistance (Li et al., 2022). To achieve the efficient rejuvenation of aged asphalt binder by bio-oil, the functionalization treatment for bio-oil is necessary, which involves purification, composition modification and component conversion, as shown in Fig. 5.

The purification procedures usually carried on bio-oil to remove moisture, light components, and residues by physical methods such as filtration, distillation, and heating (Jiang et al., 2021, 2022a; Su et al., 2018). Mirhosseini et al. (2019) prepared date seed oil by Soxhlet extraction technique to achieve the isolation and concentration of bio-oil. Gong et al. (2016) improved the purity of the bio-oil by distillation.

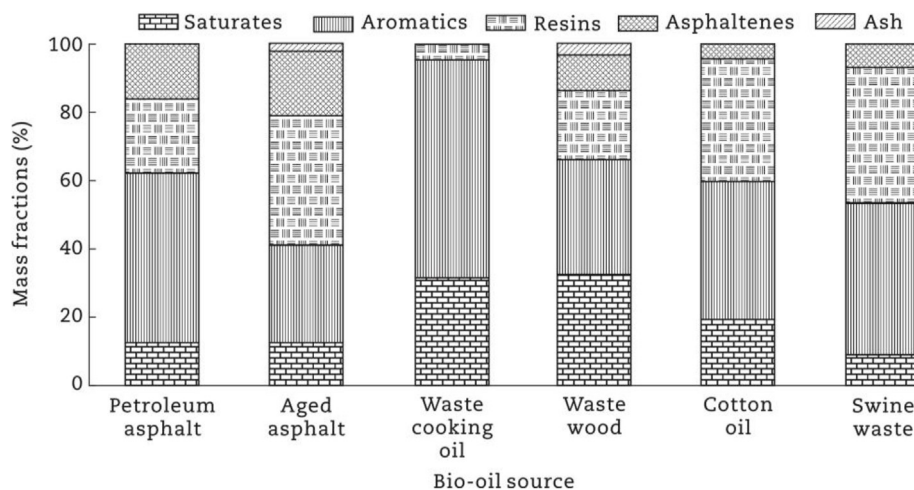


Fig. 4. Four components of petroleum asphalt and bio-oil from different sources (Bao et al., 2020; Fernández-Gómez et al., 2014; Luo and Huang, 2021; Oldham et al., 2015; Zhu et al., 2017).

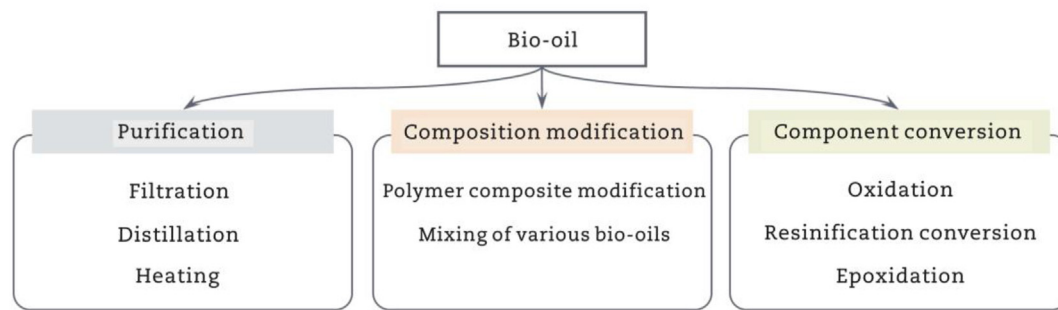


Fig. 5. Functionalization treatment methods for bio-oil.

Distillation process was used to remove a large number of polar light components of the bio-oil derived from sawdust (Zhang et al., 2017). Yu et al. (2014) heated the bio-rejuvenated asphalt rejuvenated by waste vegetable oil to remove bubbles and light components.

Since there exist some properties defects in the rejuvenation process of heavy bio-oil, the final product is commonly functionalized on the basis of concentration and purification to improve the properties. The treatment procedures can be classified into two categories (Jiang et al., 2021, 2022a; Ju et al., 2022, 2023; Su et al., 2018). (a) Polymer composite modification, a variety of bio-oil mixing modification and other methods. (b) Chemical methods such as oxidation and resinification conversion.

For the purpose of composite or mixing modification, bio-oil is usually combined with plasticizer, resin, anti-aging agent and other materials to prepare composite rejuvenators (Xu et al., 2018). Cai et al. (2019) reported that the components before and after aging were not changed significantly due to the addition of the plasticizer dibutyl phthalate (DBP) to the cotton oil. The addition of plasticizer can enhance the ability of bio-oil to soften the aged binder and improve the low-temperature cracking resistance for rejuvenated binders compared to the bio-rejuvenator without plasticizer (Zhu et al., 2017). Carrión et al. (2017) prepared the bio-material composed of pine resin (80%) and linseed oil (20%) to increase the stability of light components and to improve the rejuvenation effect. Ran et al. (2015) synthesized a bio-rejuvenator by adding antioxidants, octyl methoxycinnamate, and antistripping agent to the base oil, which improved thermal stability and aging resistance for the rejuvenated binder. Samieadel et al. (2020) reported that a balanced combination of two types of bio-oils obtained from algae and swine manure showed better rejuvenation effect than the single bio-oil.

The bio-oil produced from the liquefaction process of woody materials contains functional groups such as alcohols, esters, furfural, carboxylic acids, aromatic compounds, and their derivatives, which can be converted into large molecular aromatic compounds through condensation, grafting, and copolymerization with agents such as formaldehyde and isocyanates. In terms of chemical conversion, a few studies have reported that the liquefied product from lignocellulosic materials can be used to prepare bio-based polyurethane (PU) materials or adhesives (Cui et al., 2017; Ge et al., 2003; Zhang et al., 2006). Doh et al. (2005) prepared polymeric materials from wood-based bio-oil, which exhibits good mechanical properties. Similarly, Lee and Chen (2008) prepared thermoplastic liquefied wood-based resins. Ding et al. (2019) and Jiang et al. (2022a) analyzed the conversion process and chemical pathways of wood liquefied products (LPs) derived from straw and sawdust to bio-based phenolic resins (BPFs). These thermoplastic wood-based resins were used to improve the properties of asphalt, and the potential to be used as a direct alternative of petroleum asphalt was discussed.

In summary, the conversion of some components of bio-oil to the substances with larger molecule through functionalization procedures can effectively supplement the components missed as a result of aging, which can furthest restore the properties of the aged asphalt binders,

especially the low temperature and anti-aging properties. Therefore, based on the analysis of the composition characteristics of bio-oil, according to the decay or loss of light components during the aging process of asphalt, the functionalization conversion of bio-oil can be carried on to realize the rejuvenation of aged asphalt. However, the conversion mechanism is still unclear, and the conversion techniques and methods need further study in depth.

3. Rejuvenating behavior and mechanism of bio-oil on aged asphalt

3.1. Diffusion behavior of bio-oil into aged asphalt

The rejuvenation of aged asphalt binder by bio-oil is to restore the properties of asphalt that decay due to aging, which is not the reversal of the aging (Cavalli et al., 2018; Kaseer et al., 2019). Instead, at the molecular level, the rejuvenation of aged asphalt materials can be achieved by regulating the components and restoring the microstructure. Therefore, the physical properties, chemical composition and diffusion performance of bio-oil have a significant effect on the rejuvenator behavior of aged asphalt materials.

In the process of asphalt mixture rejuvenation, the rejuvenator first disperses on the surface of the aggregate coated by aged asphalt by mechanical stirring, and then diffuses and penetrates into the interior to rejuvenate the aged asphalt binders. Therefore, the effect of rejuvenation depends on the degree of the diffusion and the homogeneity of the asphalt binder (Cong et al., 2016; Kuang et al., 2020; Zhang et al., 2024).

At present, there are mainly two methods to characterize the diffusion and penetration behavior of rejuvenators at the interface of the aged binder. The characteristics and evaluation indicators are listed in Table 3.

The diffusion test methods are based on the diffusion model and mass transfer model, such as soaking method, layered extraction method and surface wettability method, which can reflect the diffusion behavior of rejuvenators into asphalt binder. This type of methods requires scanning electron microscopy (SEM), infrared spectroscopy, a fluorescence microscope, and other microscopic analysis technologies to characterize the properties, microstructure, and the change in component of recycled asphalt. However, there is limitation due to the high requirements for equipment, and it is very difficult to reflect the diffusion process and the homogeneity of the rejuvenated binder.

The tracking and simulation of the diffusion and penetration process of rejuvenators are another type of research method, such as tracer method and molecular dynamics simulation. The former generally uses fluorescence label to track the migration process of key components. The latter needs to determine the representative molecules of bio-oil by using Fourier transform infrared (FTIR) spectroscopy, GC-MS, gel permeation chromatography (GPC) and so on. Based on the asphalt molecular model established, the appropriate diffusion model and calculation parameters are determined, and the whole model is built to investigate the effects of bio-oil molecular morphology, aging degree, temperature and other factors on the diffusion behavior.

Table 3
Research methods of diffusion behavior of regenerants.

Characterization method of diffusion behavior		Characteristic	Evaluation index
Diffusion test method	Soaking method (Qi et al., 2014; Wang et al., 2019)	The aged asphalt sample is placed in a container containing the rejuvenator, and the state of the layered sample can reflect indirectly the diffusion process. The operation is simple, but subjected to the container.	Macro and micro evaluation indicators, diffusion coefficient
	Layered extraction method (Bowers et al., 2014; Karlsson and Isacson, 2003)	The asphalt on the surface of aggregates is extracted by using organic solvents or at the high temperature domain. It can reflect diffusion behavior, which is difficult to control the layer thickness and subjected to properties of solvents.	Macro and micro evaluation indicators
	Surface wettability method (Kuang et al., 2020)	It reflects the interface wettability behavior of rejuvenators but is not suitable for the study of penetration and diffusion behavior of rejuvenators.	Contact angle, wetting work, wetting speed, wetting time
Process tracking and simulation	Tracer method (Navaro et al., 2012)	The diffusion path of rejuvenators is tracked by using special markers. The result is visual, but easily susceptible to characteristics the tracer.	Qualitative evaluation, functional group index of tracer and other quantitative indicators
	Molecular dynamics simulation (Xu and Wang, 2018)	Based on selecting appropriate molecular models and constructing contact models, the diffusion process can be simulated from the micro-scale, and the diffusion mechanism and driving factors can also be analyzed. It is not limited to the test equipment and specimens, but subjected to models.	Diffusion coefficient

For the diffusion analysis model of rejuvenator and aged binder, it can be usually divided into layered diffusion model and point diffusion model. The layered diffusion model can be classified into two categories based on the difference of diffusion medium, one is two layered diffusion model (Xu et al., 2019), which can be used for simulating the aged asphalt and rejuvenator system. And the other is three layered diffusion model (Xu and Wang, 2018), which can simulate the diffusion and penetration of rejuvenating agent at the interface of virgin and aged asphalt binder, and it is closer to the reality. The point model system can simulate the movement process of rejuvenator dispersed in aged asphalt.

At present, a few studies have reported that bio-oil has good diffusion properties into asphalt, which can facilitate the diffusion and fusion between the new and aged asphalt binders. Wang et al. (2019) designed three diffusion tests and evaluated the diffusion efficiency of bio-oil by the sinking time, softening rate, and gravitational collapsing. The result indicated that waste cooking oil has good diffusion performance. Cuci-niello et al. (2021) compared the diffusion properties of bio-oil under different film thicknesses, and concluded that the vegetable oil had high diffusion coefficients at high temperature. For the thickness of 0.2 mm, the diffusion process of bio-oil into aged asphalt is completed in a few seconds.

Using molecular dynamics simulation, Zhang et al. (2021b) found that the diffusion coefficient of the asphaltenes and aromatics decreased as the bio-oil content increased, and an opposite result for resins was seen. Wang et al. (2022) discussed the effect of waste oil rejuvenators such as waste cooking oil, waste vegetable oil and waste engine oil on the degree of diffusion, and calculated the viscosity and solubility parameters. They found that bio-oil can facilitate the diffusion and fusion between the virgin and aged asphalt. In a study conducted by Xu et al. (2019), two separate layered diffusion model was used to simulate the diffusion behavior of rejuvenators into the long-term aging asphalt, and analyzed the driving mechanism of diffusion. The results indicated that the molecular thermal motion was not the only factor for diffusion, but the intermolecular force is another important driving factor, especially at low temperatures. Similarly, Sun and Zhou (2018) analyzed the diffusion mechanism and rheological properties of waste wood bio-rejuvenated asphalt. Ding et al. (2021) discussed the diffusion mechanism of

bio-rejuvenators into aged asphalt and established the diffusion model of rejuvenator for binder. In another study, Zhang et al. (2021c) constructed a molecular model of bio-rejuvenator for the aged asphalt, and analyzed the solubility parameter, storage stability, and morphology of the bio-rejuvenated asphalt.

In summary, the molecular size, polarity and functional groups of bio-rejuvenator significantly affect the diffusion behavior of bio-oil in aged asphalt, and bio-oil can facilitate the diffusion and fusion between the virgin and aged asphalt. The waste cooking oil and waste vegetable oil have good diffusion efficiency, and the research on the diffusion characteristic and fusion effect of bio-oil derived from lignocellulosic biomass and animal manure is insufficient. The diffusion test and tracer method can analyze the diffusion efficiency and influencing factors of rejuvenators, but it cannot reveal the diffusion process and driving mechanism of bio-oil at the molecular level. However, molecular dynamics simulation can reveal the diffusion driving factors and diffusion laws of bio-oil. It should be noted that Fick's law exhibits that diffusion rate depends on the concentration of the substance. Moreover, the concentration at this scale cannot be accurately determined, and the applicability of the law needs to be further discussed. Based on the discussion above, it is concluded that bio-oil can rejuvenate the aged binder through similarity compatibility, component penetration, migration and fusion.

3.2. Components regulation of bio-oil on aged asphalt

During the aging process, the light components generate unsaturated bonds through dehydrogenation, and then some chemical reactions occur like addition reaction. At the early stage of aging, the aromatics mainly convert into resins, and later, the resins transform to asphaltenes (Menapace and Masad, 2016). There is a certain correlation between the component change and the microstructural evolution caused by aging (Li et al., 2020). As the content of asphaltene increases in the aged asphalt, the components gradually become unbalanced, which results in a conversion of the colloidal structure of asphalt to similar gel-type structure (Behnood, 2019). Therefore, restoring the component balance is an important indicator of rejuvenation effectiveness. Researchers have carried out many studies on the component regulation of bio-oil on the

aged asphalt. The effects of various types of bio-oils on the components of aged binders are shown in Fig. 6.

As can be seen from Fig. 6, waste cooking oil, cotton oil, waste pig fat can make the content of SARA fractions of the aged base asphalt and SBS modified asphalt convert into the virgin state. This is mainly because these bio-oils are rich in light components, which can increase light components and dilute the heavy fractions. However, the asphaltene content of bio-oil derived from swine manure is slightly higher than that of the aged binder. Compared with the single bio-oil, a combination of two types of bio-oils, generating from algae and swine manure respectively, can effectively supplement light components. Therefore, swine manure-based bio-oil rich in resins cannot restore the colloidal structure of the aged binder through the diluting mechanism. Moreover, the asphaltene content in the reclaimed asphalt extracted from RAP is higher than that in the asphalt samples obtained by the aging simulation, and the aromatic content is lower, which leads to the difficulty in using a single bio-oil for the component regulation.

3.3. Asphaltene disaggregation of aged asphalt by bio-oil

Many researchers have carried out experiment investigation and numerical simulation to study the asphaltene agglomeration behavior of the aged asphalt binder and the dispersion and disaggregation of bio-oil to asphaltene. Pahlavan et al. (2018b) found that the aging exacerbates the agglomeration behavior of asphaltenes. This is because the carbonyl (C=O) groups reduce the electron distribution of the aromatic zone of asphaltenes and weaken the unfavorable repulsion between asphaltene molecules, thus facilitate the generation of asphaltene agglomerates. This agglomeration behavior of asphaltenes can result in the formation of the hard asphalt binders (Zhao et al., 2018). Some researchers have reported that the amides and polar nitrogen groups can disperse the asphaltene dimers (Fini et al., 2011; Pahlavan et al., 2018b). Thus, the bio-oil derived from swine manure mainly facilitates the dissolution of asphaltene aggregates to rejuvenate aged asphalt (Pahlavan et al., 2018a; Zadshir et al., 2018). Ding et al. (2021) pointed out that the amides and nitrogen groups can depolymerize asphaltene agglomerations through the pull-out effect and intercalation effect, which mainly depends on the molecular structure and properties of bio-oil, as shown in Fig. 7. The former can weaken the π - π interaction of asphaltene agglomerates, disaggregate partial asphaltene and interfere the

reaggregation process. The latter utilizes the interaction between bio-oil molecules and aromatic cores in asphaltenes to realize the disaggregation of the large microstructure. Samieadel et al. (2020) proposed the process of polar amide groups disaggregating asphaltenes shown as follows. (1) Bio-oil molecules were close to the asphaltene agglomerates. (2) Bio-oil molecules subsequently penetrated into the large microstructure. (3) The deagglomeration of the asphaltene microstructure was disaggregated, as shown in Fig. 8. When amides insert into asphaltene agglomerates, phenols can preferentially participate in the process (Pahlavan et al., 2019a, 2019b). In a study conducted by Pahlavan et al. (2018b), high resolution transmission electron microscopy (HRTEM) was used to verify the disagglomeration effect of bio-oil on asphaltene agglomeration.

3.4. Micro-mesoscopic rejuvenation effect of bio-oil on aged asphalt

The changes in molecular weight distribution and molecular size of bio-rejuvenated asphalt indicate that some lignocellulosic bio-oils and waste edible oils can increase light components to dilute and dissolve in the microstructure of aged asphalt (Zhao et al., 2020). Both bio-oil from swine manure and vegetable oil can effectively reduce the size of large molecules of aged asphalt (Zadshir et al., 2018). Zhao et al. (2018, 2020) utilized the GPC test and relaxation spectrum to assess rejuvenation effectiveness, and found that waste vegetable oil can reduce the content of large size molecules and increase the small size molecules. Further, they concluded that the dissolving mechanism is the interaction between hydrogen bonds and amines or hydroxyl groups of asphaltene molecules. Zadshir et al. (2018) reported that bio-oil obtained from swine manure and vegetable oil reduced the colloidal instability index and increased the stability of internal structure. It was reported that resins and aromatics played a major role in the rejuvenation process. However, due to the higher polarity of resins and much higher interaction with asphaltene molecules, resins had better rejuvenation efficacy than aromatics. The scanning electron microscopy (SEM) was used to characterize the microscopic morphologies of bio-rejuvenated asphalt in a study (Fig. 9), and the results showed that castor oil (CO) and straw oil (STO) can repair cracks on the surface of aged asphalt, and CO have better rejuvenation effects (Peng et al., 2023).

However, researchers have reported that the interaction between bio-oil and asphalt is mainly a physical fusion behavior (Cavalli et al., 2018;

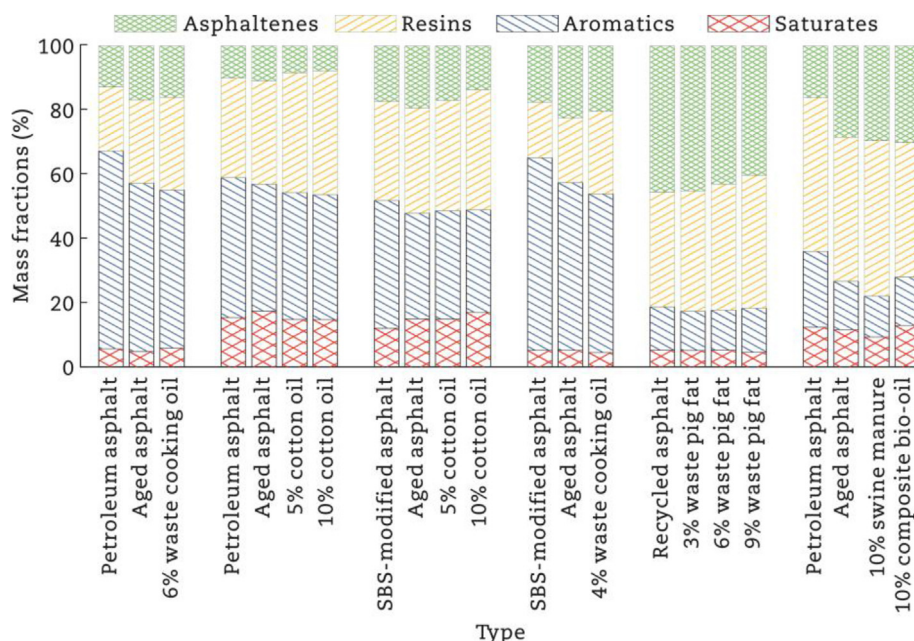


Fig. 6. Rejuvenation action of various bio-oils on aging asphalt components (Chen et al., 2014; Nciri et al., 2020; Samieadel et al., 2020; Zhu et al., 2017).

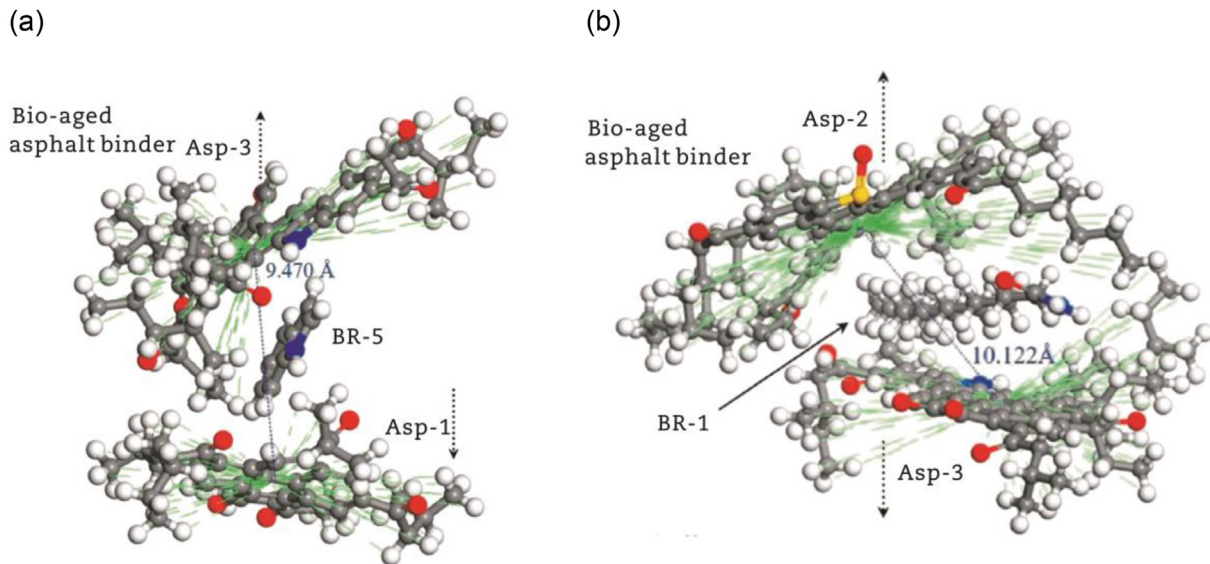


Fig. 7. Molecular dynamics simulation of the internal effect for asphaltene disintegration (Ding et al., 2021). (a) “Pull-out” effect. (b) “Intercalation” effect.

Li et al., 2019; Wang et al., 2018a). Cai et al. (2019) used FTIR tests to characterize the mechanism of rejuvenated SBS modified asphalt using cotton oil, and found that there was only a physical interaction during rejuvenating process, which cannot restore the crosslinking microstructure of SBS. The results of the FTIR tests for bio-rejuvenated and aged asphalt showed that no new functional groups were generated during the rejuvenation process, and there was no chemical interaction between the aged asphalt and bio-oil (Chen et al., 2018). Using thin-layer chromatography with flame ionization detection (TLC-FID), Gong et al. (2016) carried out chromatographic analysis of base asphalt, aged asphalt and bio-rejuvenated asphalt (Fig. 10). The results showed that the aged binder had similar chromatograms with abio-rejuvenated asphalt, indicating that the rejuvenation mechanism was not a chemical interaction. However, the content of saturates, aromatics and resins not only changed significantly, but also the chemical properties might change during the aging process.

In summary, component changes and microstructure evolution are the significant characteristics of aging process of asphalt binder. After the addition of bio-oil, the component content and molecular weight distribution of the asphalt binders can be restored to the same state as virgin asphalt, and the microstructure can be restored to various degrees. There are two main mechanisms of rejuvenation from the micro scale. The first is the dissolution of the large molecular substances such as asphaltene

agglomerates produced during the aging process. The other is the increase of maltene content and many light components diffused into the aged binder to exert dilution effect. However, these two rejuvenation mechanisms are both affected by the type of bio-oil and aged asphalt binder. From the microscopic perspective, the rejuvenation of aged binders by bio-oil from swine manure is mainly based on the dissolving mechanism, which is related to amide groups. While, the waste cooking oil and the bio-oil from lignocellulosic materials mainly dilute the aged asphalt binder, and some can restore the colloidal structure through diluting mechanism. It should be emphasized that the content of light components in bio-oil from swine manure is less, and the compatibility between amides and asphalt binders is poor. Further research is needed to focus on the rejuvenation mechanism of bio-oil on aged asphalt materials from micro scale.

4. Preparation process of bio-rejuvenated asphalt and rejuvenation effects

4.1. Preparation process of bio-rejuvenated asphalt

During the rejuvenation process of RAP, bio-oil is firstly sprayed on the surface of aged asphalt binder, and then bio-oil molecules gradually diffuse into the binder, and finally form a stable blend system. Therefore,

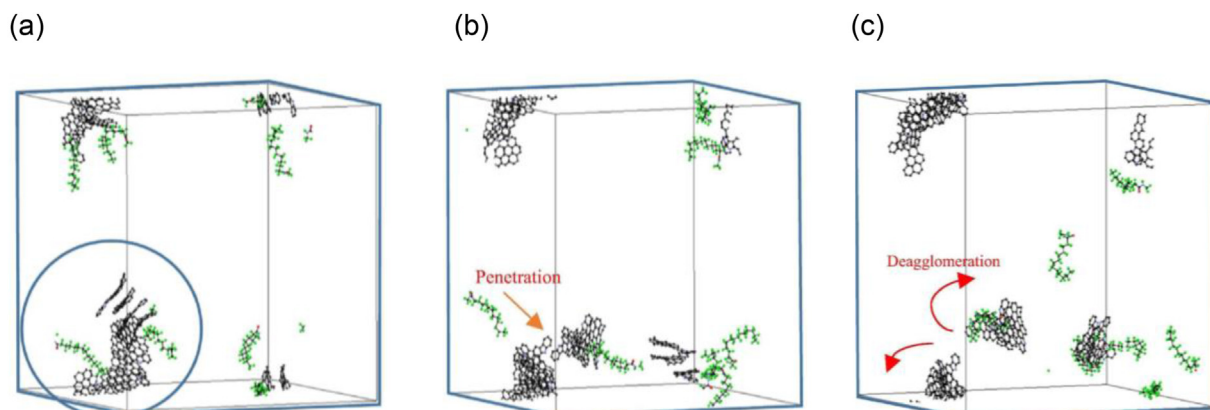


Fig. 8. Molecular dynamics simulation of disintegration procedures of asphaltene by amide-based compounds (Samieadel et al., 2020). (a) Approach. (b) Penetration. (c) Deagglomeration.

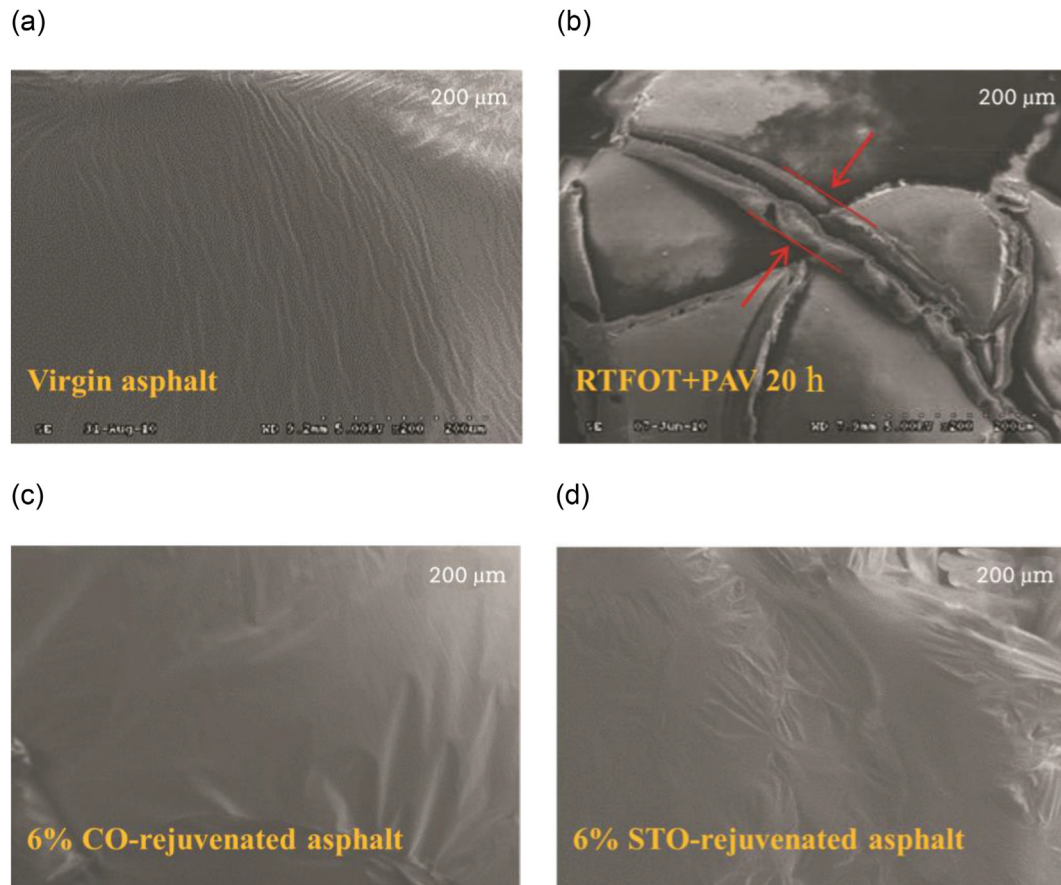


Fig. 9. SEM images of different asphalt binders (Peng et al., 2023). (a) Virgin asphalt. (b) RTFOT + PAV 20 h. (c) 6% CO-rejuvenated asphalt. (d) 6% STO-rejuvenated asphalt.

the reasonable control of process method and parameters can enhance the rejuvenation effects (Mao et al., 2021). The preparation procedure mainly has high-speed shearing and mechanical stirring. There are several process parameters such as preparation temperature, rejuvenator dosage, shear time and shear rate. The process parameters for different bio-rejuvenated asphalt are shown in Table 4, where the dose of bio-rejuvenators is by the weight of asphalt binder.

Many studies have shown that the preparation time and temperature significantly both affect the diffusion behavior of bio-oil in aged asphalt. Short blending time and low temperature will reduce the homogeneity of the final binder, while too high temperatures may result in the aging of bio-oil and rejuvenated asphalt. When the shear rate reaches a certain level, the changes in the properties of rejuvenated asphalt tend to be not

significant. Some researchers have also achieved good recovery effect by manual stirring (Zadshir et al., 2018). The determination of the rejuvenator amount is usually based on the basic performance indexes of rejuvenated asphalt, such as penetration, softening point, viscosity and ductility, which are expected to reach the same level as the virgin asphalt (Chen et al., 2014; Gong et al., 2016; Luo and Huang, 2021). In a study conducted by Gong et al. (2016), bio-oil can change the dispersion state of polymer in the rejuvenated SBS-modified asphalt and affect the recovery of the basic performance. The dosage of bio-rejuvenators varies greatly, generally less than 10% by weight of asphalt binder. Excessive content of rejuvenator can result in a significant softening of rejuvenated asphalt and even affect the performance of mixture (Li et al., 2022). In addition, the type and dosage of rejuvenator have a great effect on the

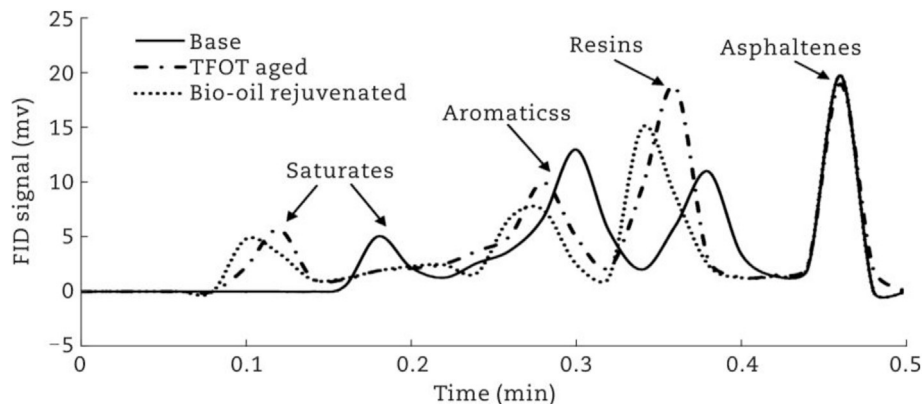


Fig. 10. Chromatographic analysis of asphalt samples (Gong et al., 2016).

Table 4

Technological parameters for various bio-rejuvenated asphalts.

Sources of bio-rejuvenator	Type of aged bitumen	Shear rate (r/min)	Shear time (min)	Shear temperature (°C)	Dose (%)	Reference
Sawdust	Petroleum asphalt	5000	15	135	10, 15, 20	Zhang et al. (2019a)
Waste soybean oil	Petroleum asphalt	200	30	130	2, 4, 6, 8, 10	Ji et al. (2017)
Cotton oil	Petroleum asphalt	5000	15	125	3, 4	Cai et al. (2019)
Swine manure	Petroleum asphalt	–	5	135	5, 10, 30	Zadshir et al. (2018)
Cotton oil	SBS modified asphalt	5000	15	160	5, 10	Zhu et al. (2017)
Waste cooking oil	SBS modified asphalt	1200	15	130	3, 4, 5, 6, 7	Chen et al. (2014)
Soybean oil	Recycled asphalt binder	2000	45	140	6, 12	Elkashaf et al. (2018)
Corn oil	Recycled asphalt binder	2000	20	140	1.5, 3, 6, 9	Zhao et al. (2018)
Waste soybean oil	Recycled asphalt binder	200	30	130	2, 4, 6, 8, 10	Ji et al. (2017)
Waste pig fat	Recycled asphalt binder	3000	120	180	3, 6, 9	Nciri et al. (2020)

fatigue resistance and low temperature properties of aged asphalt binder. Therefore, it is necessary to comprehensively consider the recovery of high temperature properties and the improvement of low temperature properties of the aged binder to determine the type and dose of bio-oil in further research.

4.2. Rejuvenating effects of bio-oil on aged asphalt

The rejuvenation of aged asphalt can be considered as the reverse process of aging. The various properties and microstructural recovery of rejuvenated asphalt have attracted the attention of many scholars. The changes in high and low temperature properties, fatigue cracking resistance, microstructure and chemical composition of bio-rejuvenated asphalt are shown in Table 5.

Table 5 shows that bio-oil can effectively restore the basic performance and rheological properties of aged asphalt, but the bio-rejuvenated asphalt is usually easier to age than virgin asphalt, which may be attributed to the high oxygen content in bio-oil. Moreover, the mass loss of bio-rejuvenated binder is greater in the course of secondary aging. This is because the bio-oil contains water and many light components such as hydroxyl, carbonyl compounds (Behnood, 2019). A few studies have reported that the anti-aging properties also significantly

depends on the type of the aged binder, and the aging resistance of rejuvenated SBS-modified asphalt is better than that of rejuvenated base asphalt (Cai et al., 2019). Thus, the secondary aging property is an important evaluation indicator to assess the durability of bio-rejuvenated asphalt and the restorative effect of bio-rejuvenator (Menapace et al., 2018). However, the secondary aging behavior and mechanism of bio-rejuvenated asphalt is still not sufficient and need further study.

Many studies have reported that the bio-rejuvenators or bio-additives can successfully restore the rheological properties of the aged asphalt binders. In a study conducted by Lin et al. (2022), based on multi-scales, the rheological properties and micro-mechanism of rejuvenated asphalt were studied, and the relationship between chemical composition, microstructure and rheological properties was analyzed. Zhao et al. (2022) investigated the effects of different components from waste cooking oil on the rheological properties and microstructure characteristics of aged asphalt. Similarly, Loise et al. (2021) discussed the effects of trimethylphenyl polyphosphate and soybean-derived oil on the rheological properties and microstructure of bio-rejuvenated asphalt by DSR tests and microscopic analysis. In a recent study conducted by Lyu et al. (2021), the rejuvenation mechanism and effects of bio-oil on the aged asphalt binders were discussed in a microscopic concept. Eltwati et al. (2022) analyzed the effects of the composite rejuvenator on the physical,

Table 5

Effect of bio-rejuvenators on properties of aged asphalt.

Performance	Test method	Bio-rejuvenator	Rejuvenation effect	Reference
High temperature properties	DSR, penetration, softening point, RV	Waste vegetable oil	Increased penetration, decreased softening point	Chen et al. (2014)
		Wood-based bio-oil	Decreased viscosity and activation energy of aged bitumen, improved flow and temperature sensitivity	Zhang et al. (2019a)
		Castor oil	Reduced rutting factor	Zeng et al., 2018
		Biodiesel residue	Decreased viscosity of rejuvenated asphalt binder and SBS-modified asphalt	Gong et al. (2016)
Intermediate temperature properties	DSR	Pine wood-based bio-oil	Increased non-recoverable creep compliance (J_{nr}), decreased percentage recovery (R), improved high temperature properties	Borghi et al. (2017)
		Corn oil, soybean oil	Improved fatigue resistance	Ji et al. (2017)
		Pine wood-based bio-oil	Increased number of cycles to fatigue failure (N_f), fatigue resistance can be restored to the same level as virgin asphalt	Borghi et al. (2017)
Low temperature properties	BBR, ductility	Swine manure	Decreased stiffness (S) and increased creep rate (m)	Oldham et al. (2018)
Anti-aging properties	DSR, FTIR, RV	Waste edible oil	Increased in ductility	Chen et al. (2014)
		Cotton oil	The rejuvenated binder was more susceptible to aging than virgin binders	Cai et al. (2019)
Temperature sensitivity	PI, PVN	Soybean oil	Compared to the neat asphalt, the rejuvenation effect of SBS modified asphalt was remarkable.	Elkashaf et al. (2017)
		Waste pig fat	An appropriate amount of bio-rejuvenator remarkably improved temperature sensitivity	Nciri et al. (2020)
Microstructure	AFM	Waste vegetable oil	Improved molecular mobility, reduced size and number of bee-structures	Yu et al. (2014)
Chemical composition	SEM	Waste cooking oil	Improved the morphology of aged asphalt surface	Li et al. (2021)
	SARA	Cotton oil	Increased saturates and aromatics	Zhu et al. (2017)
	GPC	Edible vegetable oil	The addition of bio-rejuvenator decreased the size of large molecules and which in turn increased the size of small molecules.	Zhao et al. (2020)
	FTIR	Waste vegetable oil	Decreased carbonyl (C=O) index and sulfoxide (S=O) index	Cao et al. (2018)
		Wood-based bio-oil	Decreased sulfoxide index, but the change of carbonyl index was unstable	Zhang et al. (2019a)

chemical and rheological properties of aged asphalt binders. Li et al. (2016) and Jiang et al. (2022b) analyzed the viscous flow characteristics of asphalt during aging process by using Arrhenius equation, and further discussed the aging decay behavior of bio-based resin modified asphalt binders.

A few studies have reported that the addition of bio-oil can decrease the aromatic index and the sulfoxide index, while the change trend of carbonyl index may be unstable (Behnood, 2019; Zhang et al., 2019a). Therefore, the changes in the aromatic index and the sulfoxide index can be used to evaluate the rejuvenation efficacy. However, there was no stable trend for the changes in components of the bio-oil derived from swine manure due to the high asphaltene content. It can be concluded that the properties of the bio-rejuvenated asphalt binders depend on the source and content of bio-oil, type of binders, degree of aging, and preparation procedure and process can affect the properties of the rejuvenated binder.

In summary, researchers have conducted many studies on the effects of the preparation process and composition of bio-rejuvenators on the basic performance, rheological properties and decay behavior of rejuvenated asphalt binders. However, for bio-rejuvenated asphalt, the research on the viscous flow characteristics and thermodynamic properties are still insufficient. Further research is needed to characterize and explore the viscous flow characteristics and aging dynamics related to thermodynamics to clarify the dynamic decay behavior and mechanism of the rejuvenated asphalt.

5. Conclusions and future research

To achieve effective and green rejuvenation of PAP, this review aims to present an overview on the preparation process, composition characteristics, and functionalization treatment methods for bio-oil, and then the diffusion behavior, rejuvenation mechanisms and the regeneration effect of bio-rejuvenators were discussed in this work. Based on the findings of this review, the conclusions and suggestions for future research are obtained facing the field of road engineering and material science.

- (1) Compared with physical processing and biochemistry conversion, the thermochemical conversion method provides a sustainable solution for the application of lignocellulosic materials, algae and waste animal manure in the field of road materials. High moisture biomass is not suitable for pyrolysis, but more suitable for hydrothermal liquefaction. Liquefaction technology in organic solvents requires simpler equipment and the composition of liquefied products are easy to control. It is recommended to focus on the subsequent strengthening reactions and improvement mechanism of liquefaction solvent to enhance the properties of rejuvenators.
- (2) At present, bio-oil is still difficult to replace petroleum asphalt directly, but it has good diffusion properties for penetrating into aged asphalt and can effectively dissolve the asphaltene into small-size microstructures. It has good composition correspondence with petroleum asphalt and has great diffusion and fusion effect in aged asphalt. Bio-oil is feasible to be used as sustainable rejuvenator to effectively rejuvenate the aged asphalt.
- (3) The rejuvenation mechanism of bio-oil on aged asphalt mainly includes the dissolution for the large microstructure of asphaltene and the supplement of missing maltene due to aging. Both dissolving and diluting mechanisms depend on the type of bio-oil and aged binder. Further research is suggested to explore the component mutual feedback effect and investigate the interface diffusion and fusion characteristics between bio-oil and aged asphalt.
- (4) The rejuvenation effect is not good enough for bio-oil directly used as rejuvenator, and the functionalization treatment for bio-oil can provide an effective method to develop a bio-rejuvenator with excellent performance. The functionalized bio-oil is combined with stabilizer, plasticizer, anti-aging agent and other

materials to prepare the high-performance bio-rejuvenator, which can achieve green and efficient rejuvenation of high RAP content. This is the research direction of sustainable rejuvenators, but there are still many theories and methods need to explore.

- (5) For bio-oil rejuvenated asphalt, the rheological properties and decay behavior are significantly different from those of traditional rejuvenated asphalt using petroleum-based rejuvenators. The aging behavior and decay dynamics model of bio-rejuvenated asphalt have not been well investigated in the previous studies. Therefore, the next research focus is to characterize and explore the thermodynamic viscous flow characteristics and aging dynamics of bio-rejuvenated asphalt to clarify the dynamic decay behavior and mechanism.

Declaration of competing interest

The authors do not have any conflict of interest with other entities or researchers.

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